EXP-7 Flow control at Data link layer.

- Aim:
  Write a program to implement flow control at data link layer using sliding window protocol. Simulate the flow of frames from one node to another.

- Procedure:

(i) Create sender program.
① input window size. from the user.
② Input a text message from the user.
③ Consider 1 character per frame.
④ create a frame with following field
⑤ Send the frames
⑥ Wait for acknowledgement
⑦ Reader a file called receiver buffer.
⑧ Check ACK field.

(ii) Create a receiver program.
① Read a file called sender buffer.
② check frame no
③ If frame no expected → ACK.
else NACK. in receiver_buffer

~~sender.py~~

CODE:

(5) sender.py

```
import time.
ws = input()
msg = input()
i, n = 0, len(msg)
while i<n:
    window = msg[i: i+ws]
    frams = ["%s"%i+j%s:%c" for j,c in enumerate(window)]
    with open("sender.txt", "w") as f:
        f.write("\n".join(frams))
    time.sleep(2)
try:
    with open("recive.txt") as f:
        resp = f.read().strip()
    if resp.startswith("ACK"):
        ack_no = int(resp.split(":")[1])
    else:
        print("Retransmitting window")
except FileNotFoundError:
    pass
```

(6) receive.py

```
import time
expected = 0
received_msg = []
while True:
    try:
        with open( ) as f:
            lines = [l.strip() for l in f.strip()]
```

```
except KeyboardInterrupt:
    time.sleep(0.5)
    continue
for line in lines:
    fno, char = line.split(":", 1)
    fno = int(fno)
    if fno == expected:
        received_msg.append(char)
        expected += 1
        with open() as f:
            f.write(f"ACK {expected}")
    elif fno > expected:
        with open() as f:
            f.write(f"NACK: {expected}")
            break
print("Current msg:", "".join(received_msg))
time.sleep(1)
```

- Result :

Thus the flow control mechanism at the Data link layer was successfully implemented using Sliding window Protocol.